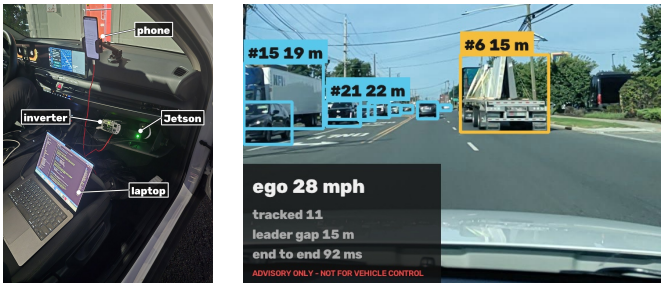
Design and On-Road Deployment of a Speed Advisory System for Preventing Traffic Jams

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(a) The rig. (b) The local layer's view on a drive.

Fig. 1. The deployed system. In (a), the phone supplies the camera, GPS and inertial unit and carries the display; the Jetson runs perception, both advisory layers and the safety filter; the inverter is its only connection to the car. In (b), the leader in the ego lane is amber and the other tracked vehicles are cyan. Nothing connects to the throttle, the brakes, the steering or the vehicle bus.

***Abstract*—Local disturbances like a burst of traffic or a stopped vehicle on a free-flowing highway can rapidly result in a traffic jam. On a road close to its operating capacity, such jams happen faster and more frequently, and take longer to drain. Preventing such jams requires a warning and intervention before they form. In this paper, we design a speed advisory system for tackling sudden traffic jams, where a commercial traffic probe feed and local sensing data is processed using an onboard compute device to identify the early signals of a forming traffic jam, and displays speed advisories to mitigate the situation before a jam fully forms. The traffic feed data is highly valuable for understanding global network dynamics but is aggregated across road segments and lags in time. The local sensing stack is real-time but limited in range. Our two layer architecture is adapted to this case, and on our simulation testbed - the Mainz highway network, prevents jams from forming in 50 of 75 cases, reduces the jam-time from 130.8 segment-minutes to 4.4 segment-minutes, and improves the throughput by 2.5% as compared to no control. We achieve this with 6.0% lower travel time in the network. We built and deployed this system in an actual car using equipment worth less than US$500, and the recorded data from driving this car was used to inform the simulation testbed.**

I. Introduction

A highway that carries traffic close to its capacity can jam without a crash or a fixed bottleneck. A short burst of inflow, or a vehicle stopped in one lane for a few minutes, pushes the local density past its critical value. The outflow of the road then falls, and the queue sustains itself, spiralling into a suddent traffic jam [1]. Such a jam also takes significantly longer to drain. We simulated such conditions on the Mainz road network [2] by blocking a lane at the exit for five minutes, and found that it increases the network's total time under jam to 137 segment-minutes. A jammed road near capacity serves less traffic than keeps arriving and the queue does not drain (Section IV). Such a jam has to be prevented rather than cleared: the warning, and the intervention, must come before the road breaks down.

Existing controllers obtain that warning from infrastructure. Variable speed limits and ramp meters hold a bottleneck below breakdown using loop detectors and overhead gantries [3], [4], and vehicle-based controllers send target speeds to equipped vehicles from a central planner that observes the traffic [5]. Most roads have neither the roadside equipment nor the coordinator. One important concern for such approaches is the cost to deployment. Installing and maintaining infrastructure costs public funds, and a central controller has maintanence and privacy costs. Adoption is the limiting factor in both cases, since adoption dictates how much public benefit gets derived from these technologies.

Our system was designed with this deployment cost and adoption in mind, with a minimum set of changes to an ordinary vehicle that can enable this advisory. We consider two sources of information: a commercial traffic probe feed for every segment of the network, and a dashboard camera. The feed is global but aggregate and late: it reports segment speeds, which barely change until a road is about to break down, commercial feeds report a change in speed a mean of 4.4 minutes after it happens [6], and the feed we use publishes once a minute and becomes readable 29 to 90 s after each publication. The camera is immediate but local: from the gaps to the vehicles ahead it measures density, which rises before speed falls, but only over the next 60 m. Thus, each modality covers what the other misses.

This paper presents a speed advisory system built on that complementarity (Fig. 2). An onboard computer in each vehicle runs two layers. The *global layer* turns each published version of the feed into a speed plan for the whole network; the plan is a deterministic function of the version, and every vehicle switches to it at a time the version itself sets, so all vehicles show the same plan without exchanging a message. The *local layer* estimates the density of the road ahead from the camera every 5 s and lowers its own vehicle's advised speed by one step below the plan when that density is about to reach its critical value. The driver sees one number: the advised speed for the current road segment.

We built the system from a phone and Jetson Nano, a US$250 onboard computer, and deployed it in a car on public roads in central New Jersey. We then built the simulated camera from the frames recorded on those drives. In simulation

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on the Mainz highway network, extended by 3 km approach roads that pace the demand into it, the system prevented the jam in 50 of 75 triggered runs, cut jam time from 130.8 to 4.4 segment-minutes per run, and served 2.5% more traffic than no control with 6.0% less travel time in the network; a controller reading the same feed without the camera left 11.6 segment-minutes of jam and served 5.4% less traffic.

Contributions. We make four contributions.

- We design a global-local advisory architecture in which vehicles agree on a network-wide plan by reading the same feed version, with no messages between them and no server computing their speeds, and each vehicle's camera corrects the plan by at most one step (Sections III-A, III-B).
- We measure the feed's delivery process and analyse the timing of feed-only control: we derive a lag condition under which no feed-only controller can act before a jam, and a sampled-margin bound that sets the local layer's horizon (Section IV).
- We deploy the system on US$500 of commodity hardware, report its latency and failure modes, and build a camera error model from 18,818 road frames (Section V).
- We evaluate the system over five jam triggers against no control and a feed-only lookahead, and test its sensitivity to camera scale, compliance and feed delay (Section VI).

II. Related Work

Holding a bottleneck below breakdown. Ramp metering [7], [4] and variable speed limits [3], [8] limit the flow that reaches a bottleneck, observing it through loop detectors and acting through roadside signals. Adding floating-car probe data and traffic camera video to such a scheme shortened jam detection and improved its speed limits in simulation [9], so a controller that learns of a jam sooner controls it better. Our system pursues the same goal without roadside equipment: the vehicles both sense and act.

Vehicles as actuators. Prior work has shown that a single controlled vehicle damps stop-and-go waves on a ring road [10], and reinforcement learning controllers have been shown to work for mixed-autonomy traffic [11], [12], [13], [14]. Our controller lies in the same vein, except the traffic state is not given but is obtained from a commercial feed and the vehicle's own camera. Earlier work has also established that coarse-grained advisories can steer human drivers [15], and the CIRCLES project deployed about one hundred vehicles on an interstate under a centralized speed planner [5]. These studies motivate our choice of a speed advisory design over alternatives. Telematics applications already show drivers feedback on their phones [16], hence, our system advises through a phone and leaves the vehicle's controls to the driver.

Probe data and coordination without messages. Speeds from GPS-equipped phones can estimate the state of a highway [17], [18], but for connected vehicles to share their own measurements over V2X [19], [20], [21], most vehicles would need to carry radios. The commercial probe feeds report a speed change minutes late [6]. To keep the decentralized design, our vehicles coordinate through common information [22], each reads the same feed version and computes the same plan from it. To mitigate the effects of lag, we use the dashboard camera to add a correction to the advisory.

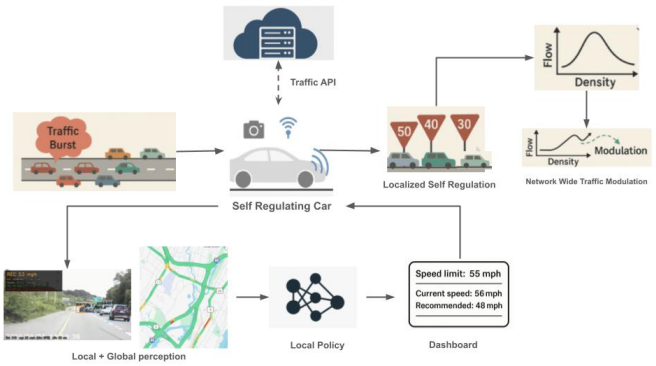


Fig. 2. System overview. A burst of inflow pushes a stretch of road toward its critical density. Each equipped vehicle reads the network from a commercial probe feed (global layer) and the road ahead from its own camera (local layer), computes its advisory onboard, and shows the driver a recommended speed. The vehicles coordinate only through the feed version they read; none talks to another.

III. Method

A. System Overview

Design goals. Five requirements shape the design. The first two are derived from the jam dynamics, and the last three are to lower the adoption cost.

(G1) *See the whole network.* A segment jams because of vehicles that entered upstream of it. Advice that prevents the jam must reach those vehicles before they arrive, so it needs the state of segments they have not yet reached.

(G2) *Act before the road breaks down.* Near capacity, a jammed bottleneck serves less traffic than keeps arriving, so the jam does not drain until demand falls (Section IV). A controller that acts only once it sees the jam must then hold traffic back for as long as demand stays high; one that acts while the road still flows can keep the jam from forming.

(G3) *Give consistent advice without a coordinator or messages between vehicles.* Vehicles on the same stretch that are told different speeds drive at different speeds, which causes lane changes and braking near capacity. A central coordinator must collect every vehicle's position and stay available, and messages between vehicles need radios and a more complicated protocol. We want the effect of coordination without the required infrastructure.

(G4) *Advise without touching the vehicle's controls.* Taking over the throttle or brakes requires modifying the vehicle and certifying the change for safety. The CIRCLES vehicles, for example, were modified to accept speed commands [5]. An advisory can be added to any car, and the driver keeps control of the vehicle.

(G5) *Run on hardware a driver can buy.* The system acts only through the vehicles that carry it. The cost of equipping one vehicle therefore limits how many vehicles can take part. The rig of Section V costs US$500.

One layer per source. No single source meets G1 and G2 together (Table I): the feed covers the network but reports

TABLE I
THE TWO SOURCES OF INFORMATION, AS MEASURED IN THIS WORK.

	Feed (global layer)	Camera (local layer)
Coverage	every network segment	60 m, own and side lanes
Measures	segment speed, jam factor	gaps, vehicles in range
Update	one version per 60 s	every 5 s
Delay to act	99.5 s	about 6 s
Sees a jam when	speed has fallen	density is rising

a forming jam late, and the camera reports it early but sees only the next 60 m. The system therefore gives each source the decision it is suited to. The global layer sets a plan for the whole network from the feed, and the local layer may lower one vehicle's speed below that plan when its own camera sees the road filling. A safety filter follows both: it raises an advisory far below the surrounding traffic, since a much slower vehicle is itself an obstruction, and withholds it when the gap ahead, or the time to close it, falls below a minimum.

Inputs and outputs. The network is divided into super-segments, twelve on the network we evaluate. For each, the global layer reads three fields from the HERE Traffic API [23], `speed`, `free_flow` and `jam_factor`, and two from the map, lane count and length. HERE does not publish how it computes the jam factor, so from the patent that defines it by travel time relative to free-flow travel time [24] we use a simple model, $10(1-v/v_f)$ clipped to $[0, 10]$, for speed v and free-flow speed v_f. The output for each super-segment is one of three fractions, 0.5, 0.75 or 1.0, of its static speed limit; we scale the static limit rather than the feed's free-flow reading so that a live reading does not enter the loop as both observation and base.

B. The Global-Local Speed Advisory

Global layer: a plan every vehicle computes the same way. Each vehicle runs the same policy π on each published version of the feed, twelve super-segments with five fields each, and obtains one speed fraction per super-segment. We trained π in simulation by Q-learning with the action set and reward of a central controller that read the full traffic state [2]: the reward pays for speed and penalises density above the critical value, refitted to our vehicles. Because π is deterministic, vehicles that read the same version compute the same plan without communicating [22]. They do not read it at the same moment, because each vehicle queries on its own schedule, so each switches to a version at a time the version itself sets (Algorithm 1). With D at least the largest delivery lag plus the query interval, every vehicle holds a version before its switch time. Querying at a common second also avoids disagreement, but a version that becomes readable during those requests, or a missed query, splits the vehicles for a full minute, and the feed's publication phase moved by 26 to 34 s during our measurements.

Local layer: a one-step correction from the camera. Every 5 s the local layer estimates the occupancy $\hat{k}$ of the road ahead, the fraction of lane length that vehicles cover: in its own lane from the gap s to the leader, as $L/(s+L)$ with $L = 4.5$ m, and in each lane beside it from the number n of vehicles seen within the camera's effective range R, as nL/R; $\hat{k}$ is the mean over the visible lanes. The layer fits the trend $\dot{k}$ of $\hat{k}$ over the last 30 s and computes the margin to the critical occupancy k_c,

$$M = \big(k_c - \hat{k} - \tau \max(\dot{k}, 0)\big)/k_c, \tag{1}$$

with $k_c = 0.178$ and horizon $\tau = 10$ s. When $M \le 0$ the vehicle shows the next lower speed fraction instead of the plan's. The correction holds for at least 30 s and releases once $M \ge 0.05$ on two consecutive samples, so one noisy frame cannot change the advice. Following Proposition 2 (§IV), τ is the layer's own response time, the sampling interval plus processing and the driver's reaction, rather than a tuned parameter.

Why the correction only lowers, and by one step. A vehicle that sees an empty road ahead does not know that the network downstream is filling, which the plan does know, so it must not speed up; a vehicle that sees a filling road needs only to slow its own arrival, and one step bounds how far it falls below its neighbours and prevents obstructing the traffic flow from the vehicle itself. Vehicles on a segment thus differ from the shared plan by at most one step, and only where their cameras see the road filling.

Algorithm 1 Version-keyed plan, run on every vehicle

1: query the feed every P_q seconds, at the vehicle's own phase
2: on a response stamped T newer than the held one: compute the plan $a = \pi(x)$
3: apply a from $T + D$, or at once if $T + D$ has passed

IV. TIMING ANALYSIS OF JAM PREVENTION

The feed publishes once a minute and arrives tens of seconds late. Every response carries a stamp, `sourceUpdated`, giving the time its version was published, and we polled the feed every 5 s for 1.9 hours over the Mainz network and over the area of our drives. Over Mainz, 113 versions arrived 60 s apart and each became readable 34.4 to 89.5 s after its stamp; over New Jersey the publication phase shifted, so versions arrived as little as 28 s apart, each readable 28.9 to 84.0 s after its stamp. The 123 responses the rig received on the road were 34 to 110 s old. A vehicle that keys its plan to the feed version (§ III-B) applies each version $D = \bar{L} + P_q$ after its stamp, where $\bar{L}$ is the largest lag and P_q the query interval: 99.5 s on the Mainz process. This delivery delay adds to the feed's own probe latency [6].

The lag condition. Let t_v be the first time a segment's speed departs from what it would have been without the triggering event, t_{on} the onset of the jam, and $W = t_{\text{on}} - t_v$ the warning the speed signal gives. Every version stamped before t_v is the same with and without the event.

Proposition 1 (Lag condition)**.** *With publications at most g seconds apart and each version applied D seconds after its stamp, a feed-only controller changes no advised speed before $t_v + D$, and so changes none before onset if $W < D$. If $D \le W < D + g$ and the stamp phase is uniform, it can act before onset with probability $(W - D)/g$.*

Proof sketch. The first stamp at or after t_v falls in $[t_v, t_v + g)$ and is applied in $[t_v + D, t_v + D + g)$; nothing applied earlier differs from the case without the event. Acting before onset needs that stamp in $[t_v, t_{\text{on}} - D)$, an interval of length $W - D$ out of g. □

The condition is necessary, not sufficient, because a changed advisory still has to change how vehicles drive before the jam forms. Speed makes W short: on the free-flow branch of the speed-density relation, speed hardly changes as density rises until the road breaks down, while density, which a camera measures from the gaps to the vehicles ahead, rises from the start.

How often the local layer must sample. A rule that observes density only at sample instants protects the critical density only if it acts early enough to absorb the rise that can happen before its next action.

Proposition 2 (Sampled margin)**.** *Let local density rise no faster than r, let a rule sample it every Δ and act l after a sample, and let its action stop the rise. If the rule acts at the first sample at or above $k^* - r(\Delta + l)$, density stays below k^*.*

Proof sketch. The sample before the one that triggers the action was below $k^* - r(\Delta + l)$, and density rises by at most $r(\Delta + l)$ before the action. □

For the feed, $\Delta + l$ is $60\,\text{s} + D$, about 160 s on the Mainz process; for a camera sampled every 5 s with about 1 s of processing it is 6 s. The local layer's margin is therefore over 25 times smaller, so it can act close to the critical density.

Why preventing a jam is worth more than draining it. A jam behaves like a queue at the bottleneck, and the queue's arithmetic decides whether it ever clears.

Proposition 3 (Point-queue drain)**.** *A queue of Q vehicles with arrival rate λ and service rate μ empties after $Q/(\mu - \lambda)$ if $\mu > \lambda$ and never otherwise. If the bottleneck serves μ_f before it breaks down and $\mu_q < \mu_f$ after, and $\mu_q \le \lambda < \mu_f$, no queue forms while it has not broken down, and none drains once it has.*

On the Mainz network the jammed exit serves 3,535 veh/h under no control, 37 veh/h less than the background demand of 3,572 veh/h, so a jam that forms under no control never drains. A controller that prevents the breakdown therefore saves more than one that only drains the queue faster, and the system is designed to act before onset.

V. Implementation and On-Road Deployment

Hardware. The rig is built from commodity parts (Fig. 1): a mid-range Android phone (OnePlus Nord N10, US$150), an NVIDIA Jetson Orin Nano developer kit (US$250 [25]), and US$100 of mount, 12 V inverter and cable, US$500 in total. Nothing connects to the vehicle's controls or its bus, and no server takes part in computing an advisory: the only network traffic is the feed query.

Software. The phone captures the camera, GPS and inertial unit, queries the feed once a minute, and forwards everything raw over USB; the Jetson runs detection (YOLOv8n at 640 px [26]), tracking, distance and density estimation, both advisory layers, the safety filter, and a scheduler that sets each sensor's sampling rate. All interpretation runs on the Jetson because placing both the sensing and compute on the phone would exacerbate the thermal problem described below.

Drives. We drove the rig on 2026-09-08 on public roads in central New Jersey. Eight runs recorded 22,929 ticks over 152.8 minutes, with a valid GPS fix on 99.1% of them, and 123 feed responses, every one returning HTTP 200. In two runs the phone applied the sampling rates the scheduler sent, and in six it logged them without applying them. No driver acted on an advisory: an experiment in which an advisory influences a driver needs institutional review board approval, which this project does not hold, and one instrumented vehicle cannot change how traffic flows, so the traffic results are simulated (§ VI).

Computation is not the bottleneck; heat is. On the Jetson the path from detection to the decoded advisory takes a median of 19.8 ms, and the phone-to-Jetson link has a 95th-percentile latency of 116.19 ms over USB against 215.63 ms over a network link, which is why the rig uses USB. The failures were physical. On the first drive the phone reached Android's severe thermal tier on a windscreen mount in sunlight and the platform closed the application; a failed cable and a hotspot shut off by sunlight cost two further drives.

From recorded frames to the simulated camera. During the drives, the frames were recorded. We ran the rig's own detector, tracker and distance code on the 18,818 frames that six drives indexed against the other sensors; the simulated camera uses the error tables of the four drives recorded at five frames per second. Measured against a second pass at 1,280 px, which sees farther, the detection rate falls from 0.95 within 10 m to 0.53 at 40 to 50 m and 0.32 at 50 to 60 m, which sets the range at 60 m. We compute distance from each box's width and an assumed vehicle width. Distance carries the frame-to-frame noise we measured, from 0.05 m at 10 m to 1.1 m at 50 to 60 m, and a per-vehicle offset with a spread of 31%, because a width-based distance is wrong by as much as that vehicle's width differs from the assumed one. The simulated camera sees its own lane and the lanes beside it within its field of view, with nearer vehicles hiding those behind them. The phone's focal length was not recorded, so the absolute distance scale is assumed, and § VI-E repeats the main result with distances scaled by 0.8 and 1.25.

VI. Evaluation in Simulation

We evaluate the system's effect on jams in simulation, with the camera model of § V; § VI-F reports separately its cost when no jam would form.

A. Testbed and Protocol

Network. We use the model of the Mainz highway network from [2] in SUMO. It has twelve super-segments and ends in an exit where we made three lanes become two, which is where it breaks down. In a simulation a burst appears at the entrances out of nowhere, while on a real road its vehicles arrive along the roads upstream, so we added a 3 km approach road, split

TABLE II
RESULTS BY TRIGGER, MEAN OVER EVALUATION SEEDS 16–30. JAM-FREE: RUNS WITH NO JAM ON THE NETWORK (OF 15). JAM: JAMMED SEGMENT-MINUTES ON THE NETWORK. THROUGHPUT AND TRAVEL TIME (IN THE NETWORK): FOR NO CONTROL, ABSOLUTE (VEH/H, VEH-H); FOR THE CONTROLLERS, THE CHANGE RELATIVE TO NO CONTROL ON THE SAME SEEDS.

Trigger	Controller	Jam-free	Jam	Throughput	Travel time
Burst 5,000	None	0	91.4	3,514	1,513
	Lookahead	15	5.3	−6.0%	+18.5%
	Global-local	15	0.1	+1.6%	−1.0%
Burst 7,500	None	0	146.9	3,513	1,683
	Lookahead	15	7.2	−4.8%	+9.6%
	Global-local	15	0.1	+2.6%	−6.9%
Burst 9,000	None	0	208.4	3,505	1,919
	Lookahead	13	16.3	−3.3%	+1.3%
	Global-local	8	12.6	+4.1%	−11.0%
Exit block	None	0	137.1	3,342	1,577
	Lookahead	0	18.4	−5.5%	+11.0%
	Global-local	0	8.1	+3.0%	−9.5%
Mid block	None	1	70.1	3,407	1,420
	Lookahead	3	11.0	−7.4%	+22.8%
	Global-local	13	1.0	+1.0%	+0.4%
All (of 75)	None	1	130.8	3,456	1,622
	Lookahead	46	11.6	−5.4%	+11.9%
	Global-local	51	4.4	+2.5%	−6.0%

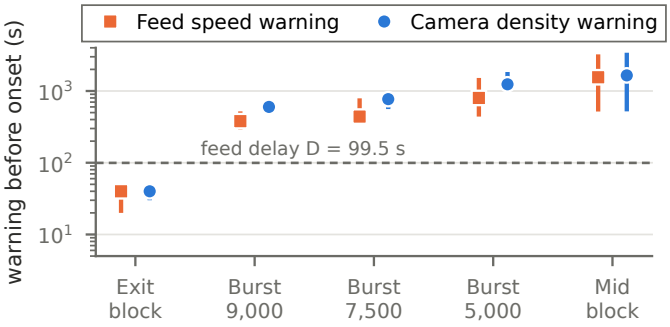


Fig. 3. How early each signal warns. For the first super-segment to jam in each no-control run (14 or 15 runs per trigger), the time before onset at which the feed's speed first fell below 80% of its pre-trigger level, and at which the camera's density estimate first rose three standard deviations above its own, each held for 30 s. Markers are medians and bars interquartile ranges; the dashed line is the feed delay D.

into three super-segments, before each of the eight entrances. The approach roads are a boundary condition that paces the demand into the network, not part of the network, and nothing on them is measured. SUMO inserts every vehicle at the far end of its approach road, earlier by that road's free-flow travel time, so that without control the network receives the same demand at the same times. A vehicle on an approach road is shown the plan of the entrance it feeds and uses its camera as on the network. Controlled vehicles follow the advice through SUMO's speed cap, so car-following still brakes for the vehicle ahead; none of the 1,305 controlled runs had a collision.

Triggers. Every run starts from a background demand of 3,572 veh/h, which lies between the flow a jammed exit serves and the flow at which a free one breaks down, so the network does not jam by itself and a jam that forms does not drain (Proposition 3). Fifteen minutes in, one of five triggers is applied. Three are bursts that raise the total inflow to

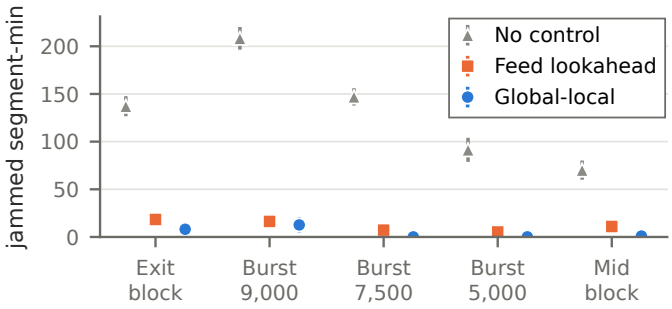


Fig. 4. Jam time by trigger, ordered by how soon no control jams: jammed segment-minutes on the network, mean and two standard errors over 15 seeds.

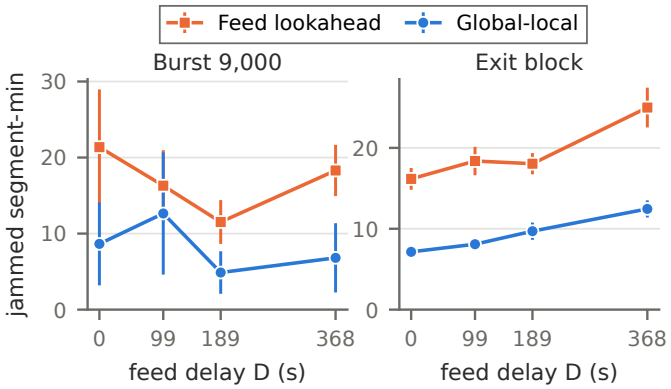


Fig. 5. Jam time against the feed delay D, mean and two standard errors over 15 seeds. No control reads no feed and is not shown; it left 208.4 and 137.1 jammed segment-minutes.

5,000 veh/h for ten minutes, or to 7,500 or 9,000 veh/h for five; they fill the network gradually, and under no control the first jam starts a median of 14 to 28 minutes after the burst begins. Two are stopped vehicles that block a road for five minutes: one lane of the exit road, which jams the exit a median of 80 s later, and two lanes mid-network, which jams it once the queue reaches the exit, a median of 37 minutes later. Each run lasts two hours, and the feed replays the polled Mainz process with $D = 99.5$ s.

Metrics. A super-segment is jammed when its mean speed is below 19 km/h, a threshold fitted to the network's speed distribution by the rule of [1], and a jam starts when that state lasts 120 s. Jam time is the jammed segment-minutes on the twelve super-segments of the network, and a run is jam-free when the network has no jam. Throughput is the flow served at the exit. Travel time is the vehicle-hours all vehicles spend in the network. Neither jam time nor travel time counts the approach roads, so the time a vehicle spends slowed on an approach road is not charged to the controller that slowed it; if a controller holds vehicles back there, the throughput at the exit falls.

Baselines and protocol. We compare the global-local advisory of § III-B with two baselines. No control lets every vehicle drive at its own desired speed. The feed lookahead is the controller one would build from the feed alone: it reads the whole network from the same delayed feed and advises 0.5 of the limit when the next super-segment downstream has

a jam factor of 2 or more, and 0.75 when the one after it has. All controllers run at full penetration on the same 15 evaluation seeds. Every threshold was chosen on five separate selection seeds, on the network without approach roads, before any evaluation seed ran. We report means over seeds and, for differences, the paired mean with two standard errors.

B. The Camera Warns Before the Feed

On the three bursts the camera's density warning came first on 43 of 45 runs, by a median of 210 to 480 s (Fig. 3). The reason is the shape of the speed-density relation: as occupancy rises to 0.05 the median segment speed falls only from 57.5 to 55.0 km/h, and then to 35.8 km/h by 0.07. On the exit block the feed's warning was shorter than D on all 15 runs, so by Proposition 1 no controller that reads only the feed could change a speed before that jam began.

C. Jams Prevented and Jam Time

Over the 75 triggered runs no control jammed on 74. The system was jam-free on 51, preventing the jam on 50 runs on which no control jammed, and cut the mean jam time from 130.8 to 4.4 segment-minutes per run, by 97% (Table II, Fig. 4). It left less jam than no control on all 15 seeds of every trigger, by 69.2 to 195.7 segment-minutes. It prevented every jam after the 5,000 and 7,500 veh/h bursts, 8 of 15 after the 9,000 veh/h burst and 12 of 14 after the mid-network block, and after the exit block, which jams within 80 s, it prevented none but shortened the jam from 137.1 to 8.1 segment-minutes.

Against the feed lookahead the system left 62% less jam over all triggers (-7.2 ± 1.7 segment-minutes), and less after four of five triggers, by 5.1 to 10.3 (at most ± 2.1). After the largest burst the difference was within noise (-3.7 ± 7.5): the lookahead was jam-free on 13 of 15 runs against the system's 8, and left 16.3 jammed segment-minutes against the system's 12.6.

D. Throughput and Travel Time

Over the triggered runs the system served 2.5% more traffic than no control ($\pm 0.3\%$), between 1.0% and 4.1% depending on the trigger, and its vehicles spent 6.0% less time in the network ($\pm 1.5\%$). Travel time in the network was below no control's after the 7,500 and 9,000 veh/h bursts and the exit block, by 6.9% to 11.0%, and within noise after the 5,000 veh/h burst and the mid-network block. The reduction is time moved onto the approach roads, which the measure does not count: the system's vehicles spent 124 vehicle-hours per run more than no control's slowed on the approach roads, more than the 97 the network saved, so counted together their travel time was 1.6% above no control's ($\pm 1.4\%$).

The lookahead served 5.4% less traffic than no control and added 11.9% to its travel time in the network, so the system served 8.3% more traffic than the lookahead and spent 16.0% less travel time in the network. The difference is what each one slows. The lookahead slows every vehicle upstream of a segment the feed reports as slowing, whether or not that vehicle's own road is filling. The local layer slows only the vehicles whose road ahead is filling, by one step, and changed the advised speed on 17% of vehicle ticks.

E. Sensitivity

Camera distance scale. We repeated the largest burst and the exit block with every camera distance scaled by 0.8 and by 1.25, since the absolute scale is assumed (§ V). On the exit block it still left 9.2 to 10.6 fewer jammed segment-minutes than the lookahead. On the burst it left 12.9 fewer at 1.25 (± 6.0), but at 0.8, which reads the road as fuller and changed the advised speed on 24% of vehicle ticks instead of 19%, it left 25.0 more (± 15.9).

Share of drivers who follow. With 75% of vehicles following, the system still left less jam than no control on both triggers. With 50% it left 29.5 more jammed segment-minutes than no control on the burst (± 24.6), though 34.5 fewer on the exit block (± 19.7). The excess comes from the local layer: with 50% following, the plan alone left 19.6 fewer than no control on the burst (± 21.0), and adding the correction raised the jam by 49.1 (± 20.2). With the feed gate of § VI-F, the excess over no control at 50% fell to 13.1 (± 24.3), within noise.

Feed delay. We varied the feed delay D (Fig. 5): 0, 99.5, 189 and 368 s, the last close to the probe latency of [6] plus the measured delivery lag. On the exit block, which jams faster than the feed reports, both controllers left more jam at the longest delay than with none, and the system stayed below the lookahead at every delay. On the largest burst the system left less jam than the lookahead at 0, 189 and 368 s, by 6.6 to 12.7 segment-minutes (at most ± 8.7), and the same at the measured delay (-3.7 ± 7.5); neither controller's jam time there rose or fell steadily with D.

F. When No Jam Would Have Formed

The local layer acts on the density ahead without knowing whether a jam would have formed, so it also slows vehicles that did not need slowing. We measured this cost separately, on runs with the background demand and no trigger, where no controller jams. The correction changed the advised speed on 15% of vehicle ticks, which reduced throughput by 6.9% (245 veh/h) and added 23.2% to the travel time in the network (108 vehicle-hours) relative to no control; the lookahead reduced throughput by 20.8% and added 26.1%.

A direct remedy is to let the local layer act only while the feed reports the network near capacity. We tested a gate of this kind: a vehicle's correction may switch on only while the feed version it applies shows a super-segment whose jam factor, measured against one step below the plan's speed for it rather than against free flow, is at least J_g. Measuring against the advised speed keeps the advisory's own slowing from holding the gate open. With $J_g = 0.5$ the correction changed the advised speed on 1.0% of vehicle ticks with no trigger, and the cost fell to 3.4% of throughput and 12.0% of travel time, close to the plan alone's 2.9% and 11.3%. After the triggers the gated system was jam-free on 56 of 75 runs and served 3.4% more traffic than no control, but it left more jam than the ungated one after the exit block ($+3.6 \pm 1.1$ segment-minutes) and the mid-network block ($+1.9 \pm 0.8$), because the gate waits for the delayed feed. The thresholds were selected on the five selection seeds by jam time, with the throughput when no trigger is applied deciding only near-ties; there the gated layer

reduced jam time less than the ungated one by more than that margin, so the results above are for the ungated layer.

VII. Conclusion and Limitations

A commercial probe feed and a dashboard camera, combined in a global-local advisory, let ordinary vehicles prevent jams that the feed alone reports too late: the feed coordinates every vehicle on one network-wide plan without messages, and the camera lowers one vehicle's speed minutes before the feed's speed would warn. The system runs on US$500 of commodity hardware, and in simulation it prevented the jam in 50 of 75 triggered runs and cut jam time by 97% while serving 2.5% more traffic than no control.

Limitations. No driver acted on an advisory, so the traffic results are simulated, on one network extended by approach roads, and they assume that most vehicles follow: with half of them following, the system left more jam than no control after the largest burst. The approach roads that pace the demand into the network are not measured, so the time vehicles spend slowed before an entrance is not charged to the system; counted, the system's travel time was 1.6% above no control's. The local layer cannot tell whether a jam would have formed, so with no trigger it reduced throughput by 6.9%; enabling it only when the feed reports the network near capacity cut that loss to 3.4% but left more jam after the exit block, which jams faster than the feed reports. The camera's errors were measured on four drives in light traffic without ground truth for distance, so its absolute scale is assumed; at a scale of 0.8 the system left more jam than the lookahead after the largest burst. The feed's delivery process was measured over two sessions on one day, and the jam factor's formula is not public.

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